Project: Worked examples
Title: Lifting lugs and spreader beams
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

This is a design calculation, not a thorough examination. Pin-loaded lifting lugs and the lug loads of spreader or lifting beams, checked by the BTH-1 allowable-stress approach (clause 3-3.3), by the EN 13155 proof route (2× the rated load without permanent set, 3× without release) or both: sling force from the rated load, sling angle and out-of-plane angle (with the automatic 6° tilt cases for a horizontal beam); tension beyond the hole, single-plane fracture beyond the hole, shear-out on two planes, pin bearing and dishing; capacity and utilisation per mode, the governing mode, the proof load and a marking and certification checklist. Scope notes (LOLER and PUWER): this sheet does not replace the thorough examination and inspection of lifting equipment by a competent person (LOLER reg 9), the marking, test and certification that the regulations and the standards require, the planning of the lifting operation, the weld, bolt and beam checks of the linked sheets, or the pin, shackle and sling ratings. It is available only while the lifting feature is enabled; its issue needs a competent-person attestation and a second independent check. Signatures are not authenticated by TempestOS: the signed-in names of the author, the checkers and the competent person are as entered into this installation.

WHAT IT CALCULATES	RELATION
Sling force	$F_v = WLL/n_{eff}$ ($n_{eff} = n_{lugs}$, at most 3); $F_s = F_v/(\cos\theta\cdot\cos\beta)$; $F_{ip} = F_s\cdot\cos\beta$; $F_{op} = F_s\cdot\sin\beta$; $N_{beam} = F_v\cdot\tan\theta$; $M_{op} = F_{op}\cdot h$
Horizontal beam, automatic 6° tilt	pitch ($\theta \pm 6°$, load redistributed, the sling at $\theta - 6°$ takes $2\cdot F_v\cdot\sin(\theta + 6°)/\sin 2\theta$, and the force along the tilted beam is $F_v\cdot\sin(\theta + 6°)/\cos\theta$); roll ($\beta + 6°$); the larger F_s governs, F_{ip}, F_{op} and N_{beam} are the envelopes
Lug section geometry	side = $\min(b_e, R - D_h/2)$; $b_{e,eff} = \min(\text{side}, 4t, 0.6\cdot(f_u/f_y)\cdot\sqrt{(D_h/\text{side})}\cdot\text{side})$; $a_{eff} = a$ (default $R - D_h/2$); $A_t = 2\cdot t\cdot b_{e,eff}$; $A_{fr} = [1.13\cdot a_{eff} + 0.92\cdot\text{side}/(1 + \text{side}/D_h)]\cdot t$; $\varphi = 55\cdot D_p/D_h$ (degrees); $A_v = 2\cdot t\cdot[a_{eff} + (D_p/2)(1 - \cos\varphi)]$; $A_b = D_p\cdot t$
Pin clearance	$C_r = 1 - 0.275\cdot\sqrt{(1 - D_p^2/D_h^2)}$
BTH-1 (3-3.3)	$F_a = \min(f_y/n_y, f_u/n_u)$; design load $D = N_d\cdot k_{svc}\cdot F_s$; $P_t = C_r\cdot F_a\cdot A_t$; $P_{sp} = C_r\cdot F_a\cdot A_{fr}$; $P_v = F_a\cdot A_v/\sqrt{3}$; $P_b = k_b\cdot F_a\cdot A_b$; $u = D/P$ for each; dishing $D_h/t \le \text{dish}_{max}$
EN 13155 (2× elastic, 3× no release)	$P_{EN} = \min(Y/2, U/3)$ for each section, Y with f_y and U with f_u (shear with $f/\sqrt{3}$; fracture area without C_r); $u = F_s/P_{EN}$
Lug capacity and proof load	$WLL_{cap} = WLL/u_{lug}$ for the governing section of each mode (smaller of the modes used); beam: u_{beam} from N_{beam} on A_{beam}; proof load = $k_{proof}\cdot WLL$

Basis: Pin-connected plates (design of lifting lugs)[1](3-3.3); Non-fixed load lifting attachments: 2× elastic, 3× no release[2](design requirements (clause to be confirmed)); Proof of static strength: the calculation route cited, not implemented[3](5); Marking of lifting equipment; thorough examination and inspection[4](reg 7, reg 9); Suitability and maintenance of work equipment[5](reg 4, reg 5).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Design mode	both		assumption: Both modes are run so the sheet shows which governs; the owner's code route decides which counts.	any
WLL	Rated load (working load limit) of the appliance, carried by all its lugs	150	kN	spec: Lifting beam specification LB-150, rated load 150 kN	0.5 … 10000 kN
n_{lugs}	Number of lugs sharing the load (a fourth is not relied on: three share)	2		drawing: LB-150-01 rev A	any
θ	Sling angle θ from the vertical, measured in the plane of the lug (the out-of-plane angle β is applied on top: $F_s = F_v/(\cos\theta\cdot\cos\beta)$)	30	°	drawing: LB-150-01 rev A	0 … 60 °
β	Out-of-plane angle β of the sling out of the lug plane	2	°	assumption: Allowance for the sling not hanging exactly in the plane of the lug.	0 … 30 °
beam	The lugs are on a horizontal spreader or lifting beam (adds the automatic 6° pitch and roll tilt cases)	yes		drawing: LB-150-01 rev A	any
n_{cycles}	Load cycles in the life of the lug (the 2×/3× route covers up to 16,000; beyond that a fatigue assessment is needed and the sheet does not run)	1000		spec: Lifting beam specification LB-150: not more than 1000 lifts in its life	1 … 16000
t	Lug plate thickness t	25	mm	drawing: LB-150-02 rev A	5 … 150 mm
Dh	Hole diameter D_h	55	mm	drawing: LB-150-02 rev A	10 … 500 mm
Dp	Pin diameter D_p (not larger than the hole)	50	mm	drawing: LB-150-02 rev A	8 … 500 mm
R	Lug end radius R about the hole centre (the plate is 2R wide at the hole)	75	mm	drawing: LB-150-02 rev A	15 … 600 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
be	Edge distance b_e from the hole edge to the plate edge on each side, across the load line (limited to R − D_h/2; the effective width for tension is reduced further by 4t and f_u/f_y)	40	mm	drawing: LB-150-02 rev A	5 … 400 mm
h	Distance h from the hole centre to the weld or base, for the out-of-plane moment (blank: no moment given)	180	mm	drawing: LB-150-02 rev A	1 … 3000 mm
f_y	Yield strength f_y of the lug plate for its thickness	345	MPa	spec: S355 plate 25 mm, mill certificate MC-2291 (16 mm < t ≤ 40 mm)	100 … 1100 MPa
f_u	Ultimate strength f_u of the lug plate	470	MPa	spec: S355 plate 25 mm, mill certificate MC-2291 (lower bound of the ultimate range)	150 … 1400 MPa
Nd	Design factor N_d of the Design Category (BTH-1 mode; user record)	1.8		assumption: Illustrative user-record value for this worked example, not a value from any standard: replace with the design factor of your Design Category from your own copy of the code.	1 … 5
n_y	Divisor n_y on f_y; the allowable stress is F_a = min(f_y/n_y, f_u/n_u) (BTH-1 mode; user record)	1.6		assumption: Illustrative user-record value, not from a standard: replace with your own record.	1 … 4
n_u	Divisor n_u on f_u; the allowable stress is F_a = min(f_y/n_y, f_u/n_u) (BTH-1 mode; user record)	2.2		assumption: Illustrative user-record value, not from a standard: replace with your own record.	1 … 6

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
k_b	Bearing multiple k_b of the allowable stress on the projected pin area (BTH-1 mode; user record)	1.2		assumption: Illustrative user-record value, not from a standard: replace with your own record.	0.5 ... 3
$dish_{max}$	Dishing-ratio limit on D_h/t (blank: the dishing check is not made; user record)	40		assumption: Illustrative user-record value, not from a standard: replace with your own record.	1 ... 100
k_{proof}	Proof-load factor on the rated load (blank: no proof load given; user record)	1.5		assumption: Illustrative user-record value, not from a standard: replace with the proof-load factor of your procedure.	1 ... 3
A_{beam}	Gross cross-section area of the spreader or lifting beam, for its axial utilisation (blank: lug utilisation only)	80	cm²	drawing: LB-150-03 rev A	1 ... 50000 cm²

3. Assumptions

A1. A design calculation, not a thorough examination: it checks the lug plate at the pin hole against the user's chosen criteria and nothing else. (LOLER reg 9 and PUWER duties are the owner's and the competent person's.)
A2. Nominal-stress screening of four sections (tension beyond the hole, single-plane fracture beyond the hole, shear-out, bearing) on a plate loaded through a pin; no stress-concentration factor, no fatigue (the cycle count is limited to 16,000), no weld, bolt or beam check, which are made on the linked sheets. (The notch-stress route of EN 13001-3-1 is cited, not implemented.)
A3. The load on the hole is the resultant sling force F_s whatever the out-of-plane angle, which is conservative; the out-of-plane component and its moment are outputs for the weld and base checks. ($F_{op} = F_s \cdot \sin \beta$.)
A4. Equal sharing between the lugs, with a fourth lug not relied on (three share). The share of a statically indeterminate set is not computed. (Conservative convention; replace with the actual share if it is known.)
A5. A horizontal beam adds the automatic 6° tilt as two cases: pitch (sling angles $\theta \pm 6°$ with the load redistributed between the lugs by statics, the slings meeting at the hook) and roll (out-of-plane angle + 6°), the larger sling force governing. N_{beam} is the force along the beam axis: in the pitch case each sling stays at θ to the beam's normal, so the more loaded lug sends $F_v \cdot \sin(\theta + 6°)/\cos \theta$ along the beam. With $\theta \le 6°$ the pitch case puts the whole load on one lug, with $W \cdot \sin 6°$ along the beam; with more than two lugs the two-lug redistribution is applied to each lug's share. (Owner's specification (6° tilt of a horizontal beam); EN 13155.)
A6. The allowable stress is the one lesser value $\min(f_y/n_y, f_u/n_u)$ for every BTH-1 section, as in the published worked example the sheet is checked against; the pin-clearance factor C_r is applied as stated from $D_p/D_h < 1$ with no cut-off. (Conservative; owner checklist.)
A7. Signatures are not authenticated by TempestOS: the platform asks for no password, so the author, the two checkers and the competent person are as entered into this installation; any person can switch to

another name. (ADR-0173; the feature stays off until this is addressed.)
A8. The design factor, service-class factor, allowable-stress divisors, bearing multiple, dishing limit and proof-load factor are the user's records from their own licensed copy of the code; none is shipped and none has a default. (ASME BTH-1 and EN 13155 text and tables are not shipped.)
A9. The equations are written out in the spec document in the author's notation (the single-plane fracture and effective-width relations and the 1.13 and 0.92 constants follow the published course example); the owner compares them with BTH-1 (2023 or later) and EN 13155+A1 before the feature is ever enabled. (Owner test pack, S8 review checklist.)
A10. mode (Design mode) is taken as both. (Both modes are run so the sheet shows which governs; the owner's code route decides which counts.)
A11. β (Out-of-plane angle β of the sling out of the lug plane) is taken as 2 °. (Allowance for the sling not hanging exactly in the plane of the lug.)
A12. Nd (Design factor N_d of the Design Category (BTH-1 mode; user record)) is taken as 1.8. (Illustrative user-record value for this worked example, not a value from any standard: replace with the design factor of your Design Category from your own copy of the code.)
A13. n_y (Divisor n_y on f_y; the allowable stress is $F_a = \min(f_y/n_y, f_u/n_u)$ (BTH-1 mode; user record)) is taken as 1.6. (Illustrative user-record value, not from a standard: replace with your own record.)
A14. n_u (Divisor n_u on f_u; the allowable stress is $F_a = \min(f_y/n_y, f_u/n_u)$ (BTH-1 mode; user record)) is taken as 2.2. (Illustrative user-record value, not from a standard: replace with your own record.)
A15. k_b (Bearing multiple k_b of the allowable stress on the projected pin area (BTH-1 mode; user record)) is taken as 1.2. (Illustrative user-record value, not from a standard: replace with your own record.)
A16. $dish_{max}$ (Dishing-ratio limit on D_h/t (blank: the dishing check is not made; user record)) is taken as 40. (Illustrative user-record value, not from a standard: replace with your own record.)
A17. k_{proof} (Proof-load factor on the rated load (blank: no proof load given; user record)) is taken as 1.5. (Illustrative user-record value, not from a standard: replace with the proof-load factor of your procedure.)

4. Working

[19] $n_{eff} = \min(n_{lugs}, 3)$ [6]
$= \min(2, 3)$
$= 2$
Lugs sharing the load (a fourth is not relied on)

[20] $F_v = WLL/n_{eff}$ [6]
$= 150\ kN / 2$
$= 75\ kN$
Vertical load on each lug

[21] $F_{s,nom} = F_v/(\cos\theta \cdot \cos\beta)$ [6]
$= 75\ kN / (\cos 30° \times \cos 2°)$
$= 86.655\ kN$
Sling force on each lug, untilted

[22] $F_{s,pitch} = 2 \cdot F_v \cdot \sin(\theta + 6°)/\sin 2\theta / \cos\beta$ [2]
$= 2 \times 75\ kN \times \sin(36°) / \sin(60°) / \cos(2°)$
$= 101.87\ kN$
Sling force on the more loaded lug when the beam pitches 6°

[23] $F_{s,roll} = F_v/(\cos\theta \cdot \cos(\beta + 6°))$ [2]
$= 75\ kN / (\cos 30° \times \cos 8°)$
$= 87.454\ kN$
Sling force when the beam rolls 6°

[24] $F_s = \max(F_{s,nom}, F_{s,pitch}, F_{s,roll})$ [6]
$=$ governing case: pitch
$= 101.87\ kN$
Sling force on each lug (resultant), governing case

[25] F_{ip} = max over the cases of $F_s \cdot \cos \beta$ [6]
= 101.87 kN × cos β
= 101.81 kN
In-plane component of the sling force (envelope)
[26] F_{op} = max over the cases of $F_s \cdot \sin \beta$ [6]
= 101.87 kN × sin β
= 12.171 kN
Out-of-plane component of the sling force (envelope)
[27] $N_{beam} = \max(F_v \cdot \tan \theta,\ F_v \cdot \sin(\theta + 6°)/\cos \theta)$ [6]
= 50.904 kN, the larger of the level and the pitched case
= 50.904 kN
Axial force along the spreader beam at the more loaded lug (envelope over the cases)
[29] $M_{op} = F_{op} \cdot h$ [6]
= 12.171 kN × 180 mm
= 2.1908 kN·m
Out-of-plane moment at the weld or base
[30] $side = \min(b_e,\ R - D_h/2)$ [6]
= min(40, 75 - 55/2) mm
= 40 mm
Edge distance on each side of the hole
[31] a_{eff} = a, or $R - D_h/2$ [6]
= 75 - 55/2 mm
= 47.5 mm
End distance along the load line
[32] $\varphi = 55 \cdot D_p/D_h$ (degrees) [1](3-3.3)
= 55 × 50 / 55
= 50 deg
Angle of the shear planes
[33] $be_{eff} = \min(b_{e,eff},\ 4t,\ 0.6 \cdot (f_u/f_y) \cdot \sqrt{(D_h/b_{e,eff})} \cdot b_{e,eff})$ [1](3-3.3)
= min(40, 100, 38.339) mm
= 38.339 mm
Effective width beyond the hole in tension
[34] $A_t = 2 \cdot t \cdot b_{e,eff}$ [1](3-3.3)
= 2 × 25 × 38.339 mm
= 1917 mm²
Area beyond the hole in tension
[35] $A_{fr} = (1.13 \cdot a + 0.92 \cdot b_e/(1 + b_e/D_h)) \cdot t$ [1](3-3.3)
= (1.13 × 47.5 + 0.92 × 40 / (1 + 40/55)) × 25 mm
= 1874.5 mm²
Single-plane fracture area beyond the hole (before C_r)
[36] $A_v = 2 \cdot t \cdot [a + (D_p/2)(1 - \cos \varphi)]$ [1](3-3.3)
= 2 × 25 × [47.5 + (50/2) × (1 - cos 50°)] mm
= 2821.5 mm²
Shear area of the two shear-out planes
[37] $A_b = D_p \cdot t$ [6]
= 50 × 25 mm
= 1250 mm²
Projected bearing area of the pin
[38] $Cr = 1 - 0.275 \cdot \sqrt{(1 - D_p^2/D_h^2)}$ [1](3-3.3)
= 1 - 0.275 × √(1 - (50/55)²)
= 0.88544
Pin-clearance factor

[39] $F_a = \min(f_y/n_y,\ f_u/n_u)$ [6]
$= \min(345 / 1.6,\ 470 / 2.2)\ \text{MPa}$
$= 213.64\ \text{MPa}$
Allowable stress

[40] $D_{bth} = N_d \cdot k_{svc} \cdot F_s$ [1](3-3.3)
$= 1.8 \times 1 \times 101.87\ \text{kN}$
$= 183.37\ \text{kN}$
Design load on each lug

[41] $P_t = C_r \cdot F_a \cdot A_t$ [1](3-3.3)
$= 0.88544 \times 213.64\ \text{MPa} \times 1917\ \text{mm}^2$
$= 362.61\ \text{kN}$
Tension beyond the hole

[42] $P_{sp} = C_r \cdot F_a \cdot A_{fr}$ [1](3-3.3)
$= 0.88544 \times 213.64\ \text{MPa} \times 1874.5\ \text{mm}^2$
$= 354.58\ \text{kN}$
Single-plane fracture beyond the hole

[43] $P_v = F_a \cdot A_v/\sqrt{3}$ [1](3-3.3)
$= 213.64\ \text{MPa} \times 2821.5\ \text{mm}^2\ /\ \sqrt{3}$
$= 348.01\ \text{kN}$
Double-plane shear-out

[44] $P_b = k_b \cdot F_a \cdot A_b$ [1](3-3.3)
$= 1.2 \times 213.64\ \text{MPa} \times 1250\ \text{mm}^2$
$= 320.45\ \text{kN}$
Pin bearing

[45] $u_t = D/P$ [1](3-3.3)
$= 183.37\ \text{kN} / 362.61\ \text{kN}$
$= 0.50568$
Utilisation, tension beyond the hole

[46] $u_{sp} = D/P$ [1](3-3.3)
$= 183.37\ \text{kN} / 354.58\ \text{kN}$
$= 0.51713$
Utilisation, single-plane fracture

[47] $u_v = D/P$ [1](3-3.3)
$= 183.37\ \text{kN} / 348.01\ \text{kN}$
$= 0.52689$
Utilisation, double-plane shear

[48] $u_b = D/P$ [1](3-3.3)
$= 183.37\ \text{kN} / 320.45\ \text{kN}$
$= 0.5722$
Utilisation, bearing

[49] $u_{bth} = \max(u_t,\ u_{sp},\ u_v,\ u_b)$ [1](3-3.3)
$=$ governing BTH-1 section
$= 0.5722$
BTH-1 utilisation

[50] $cap_{bth} = WLL/u_{bth}$ [1](3-3.3)
$= 150\ \text{kN} / 0.5722$
$= 262.14\ \text{kN}$
Rated load the lug carries by the BTH-1 approach

[52] $dish = D_h/t$ [6]
$= 55 / 25$
$= 2.2$
Dishing ratio

[53] $u_{dish} = (D_h/t)/dish_{max}$ 6
$= 2.2 / 40$
$= 0.055$
Dishing-ratio utilisation
[54] $P_{en,t} = \min(Y/2, U/3)$ 2
$= \min(661.35\ kN / 2, 900.97\ kN / 3)$
$= 300.32\ kN$
EN 13155 capacity, tension beyond the hole
[55] $u_{en,t} = F_s/P_{EN}$ 2
$= 101.87\ kN / 300.32\ kN$
$= 0.3392$
EN 13155 utilisation, tension beyond the hole
[56] $P_{en,sp} = \min(Y/2, U/3)$ 2
$= \min(646.7\ kN / 2, 881.02\ kN / 3)$
$= 293.67\ kN$
EN 13155 capacity, single-plane fracture
[57] $u_{en,sp} = F_s/P_{EN}$ 2
$= 101.87\ kN / 293.67\ kN$
$= 0.34688$
EN 13155 utilisation, single-plane fracture
[58] $P_{en,v} = \min(Y/2, U/3)$ 2
$= \min(562.01\ kN / 2, 765.63\ kN / 3)$
$= 255.21\ kN$
EN 13155 capacity, double-plane shear
[59] $u_{en,v} = F_s/P_{EN}$ 2
$= 101.87\ kN / 255.21\ kN$
$= 0.39916$
EN 13155 utilisation, double-plane shear
[60] $P_{en,b} = \min(Y/2, U/3)$ 2
$= \min(431.25\ kN / 2, 587.5\ kN / 3)$
$= 195.83\ kN$
EN 13155 capacity, bearing
[61] $u_{en,b} = F_s/P_{EN}$ 2
$= 101.87\ kN / 195.83\ kN$
$= 0.52018$
EN 13155 utilisation, bearing
[62] u_{en} = max over the four sections 2
= governing EN 13155 section
$= 0.52018$
EN 13155 utilisation
[63] $cap_{en} = WLL/u_{en}$ 2
$= 150\ kN / 0.52018$
$= 288.36\ kN$
Rated load the lug carries by the EN 13155 route
[64] lug_{util} = max over the modes used 6
= governing lug section
$= 0.5722$
Governing lug utilisation
[65] $capacity = WLL/lug_{util}$ 6
$= 150\ kN / 0.5722$
$= 262.14\ kN$
Rated load the lug carries in the governing mode

[67] $u_{beam} = \max(N_d \cdot k_{svc} \cdot N_{beam}/(A \cdot F_a),\ \max(2N/(A \cdot f_y),\ 3N/(A \cdot f_u)))$ [6]
$= N_{beam} = 50.904\ kN \text{ on } 80\ cm^2$
$= 0.053611$
Axial yield utilisation of the beam (buckling: S7)

[68] $util = \max(lug_{util},\ u_{beam},\ u_{dish})$ [6]
$= \max(0.5722,\ 0.053611,\ 0.055)$
$= 0.5722$
Governing utilisation of the lug and the spreader

[70] $proof_{load} = k_{proof} \cdot WLL$ [6]
$= 1.5 \times 150\ kN$
$= 225\ kN$
Proof-load value

5. Results

SYMBOL	VALUE	DESCRIPTION
capacity	262.14 kN	Rated load the lug carries in the governing mode
$governing_{mode}$	"BTH-1: bearing"	Governing mode and section
util	0.57	Governing utilisation of the lug and the spreader
F_s	101.87 kN	Sling force on each lug
F_{ip}	101.81 kN	In-plane component
F_{op}	12.17 kN	Out-of-plane component
N_{beam}	50.90 kN	Horizontal component per lug
cap_{bth}	262.14 kN	Capacity, BTH-1 mode
cap_{en}	288.36 kN	Capacity, EN 13155 mode
M_{op}	2.19 kN·m	Out-of-plane moment
$F_{s,pitch}$	101.87 kN	Sling force, beam pitched 6°
$F_{s,roll}$	87.45 kN	Sling force, beam rolled 6°
side	40.0 mm	Edge distance on each side of the hole
be_{eff}	38.3 mm	Effective width beyond the hole
a_{eff}	47.5 mm	End distance along the load line
Cr	0.88544	Pin-clearance factor C_r
φ	50.00°	Shear-plane angle φ
A_v	2821.5 mm²	Shear area
A_t	1917.0 mm²	Tension area beyond the hole
A_{fr}	1874.5 mm²	Fracture area beyond the hole
A_b	1250.0 mm²	Bearing area
Fa	213.6 MPa	Allowable stress
D_{bth}	183.37 kN	Design load on each lug
P_t	362.61 kN	Tension beyond the hole
P_{sp}	354.58 kN	Single-plane fracture

SYMBOL	VALUE	DESCRIPTION
P_v	348.01 kN	Double-plane shear
P_b	320.45 kN	Bearing
dish	2.2	Dishing ratio
$P_{en,t}$	300.32 kN	EN 13155 capacity, section t
$P_{en,sp}$	293.67 kN	EN 13155 capacity, section sp
$P_{en,v}$	255.21 kN	EN 13155 capacity, section v
$P_{en,b}$	195.83 kN	EN 13155 capacity, section b
$proof_{load}$	225.00 kN	Proof-load value
$sheet_{notice}$	"This is a design calculation, not a thorough examination. It does not replace the thorough examination by a competent person (LOLER reg 9) or the marking, testing and certification that the regulations require. Signatures are not authenticated by TempestOS: the signed-in names of the author, the checkers and the competent person are as entered into this installation."	Status of this sheet
checklist	"(1) mark the appliance with its rated load and a unique identification (LOLER reg 7); (2) thorough examination by a competent person before first use and at the intervals required (LOLER reg 9) — this sheet is not one; (3) proof load 225 kN, test certificate where the route requires it; (4) material certificates for the lug plate and the pin, and weld procedure and inspection records for the lug …	Marking and certification checklist
lug_{util}	0.57	Governing lug utilisation
u_{bth}	0.57	Utilisation, BTH-1 mode

SYMBOL	VALUE	DESCRIPTION
u_{en}	0.52	Utilisation, EN 13155 mode
u_{beam}	0.05	Beam axial utilisation
u_t	0.51	Utilisation, tension
u_{sp}	0.52	Utilisation, fracture
u_v	0.53	Utilisation, shear
u_b	0.57	Utilisation, bearing (governing)
u_{dish}	0.06	Dishing utilisation
$u_{en,t}$	0.34	EN 13155 utilisation, section t
$u_{en,sp}$	0.35	EN 13155 utilisation, section sp
$u_{en,v}$	0.40	EN 13155 utilisation, section v
$u_{en,b}$	0.52	EN 13155 utilisation, section b

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	BTH-1: bearing	$u_b = 0.57$	$u_b \le 1$	0.57	[1](3-3.3)
PASS	BTH-1: double-plane shear	$u_v = 0.53$	$u_v \le 1$	0.53	[1](3-3.3)
PASS	EN 13155 (3×WLL no release): bearing	$u_{en,b} = 0.52$	$u_{en,b} \le 1$	0.52	[2]
PASS	BTH-1: single-plane fracture	$u_{sp} = 0.52$	$u_{sp} \le 1$	0.52	[1](3-3.3)
PASS	BTH-1: tension beyond the hole	$u_t = 0.51$	$u_t \le 1$	0.51	[1](3-3.3)
PASS	EN 13155 (3×WLL no release): double-plane shear	$u_{en,v} = 0.40$	$u_{en,v} \le 1$	0.40	[2]
PASS	EN 13155 (3×WLL no release): single-plane fracture	$u_{en,sp} = 0.35$	$u_{en,sp} \le 1$	0.35	[2]
PASS	EN 13155 (3×WLL no release): tension beyond the hole	$u_{en,t} = 0.34$	$u_{en,t} \le 1$	0.34	[2]
PASS	BTH-1: dishing ratio	$u_{dish} = 0.06$	$u_{dish} \le 1$	0.06	[6]

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Spreader beam axial yield	$u_{beam} = 0.05$	$u_{beam} \leq 1$	0.05	[6]

7. Applicability

WARNING records: tabulated values not taken from a reference record are f_y, f_u (Material properties); Nd (Design factor and service-class factor); n_y, n_u, k_b (Allowable-stress divisors and bearing multiple); $dish_{max}$, k_{proof} (Dishing limit and proof-load factor). Enter each as one so its source, edition and licence travel with it. [record]
WARNING k_{svc}: no service-class factor entered, 1.0 used; enter it from your own copy of the code if it applies. [k-svc]
WARNING beam: the automatic 6° tilt cases are applied (pitch: sling angles θ ± 6° with the load redistributed between the lugs; roll: out-of-plane angle + 6°) and the larger sling force governs. [tilt]
WARNING be: the effective width 38.339 mm is below the edge distance 40 mm (limited to 4t and by f_u/f_y), so the tension area uses the smaller. [be-eff]

8. Verification

```
Template struct.lifting@1.1.1 Content hash 2d4633b8b90f3036 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. American Society of Mechanical Engineers. Design of Below-the-Hook Lifting Devices. ASME BTH-1. ASME.
2. BS EN 13155.
3. BS EN 13001-3-1.
4. LOLER 1998.
5. PUWER 1998.
6. spec.